Multilevel Determinants of Player Market Value in Elite European Football

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1 Introduction

The European football transfer market has evolved from a subjective scouting exercise into a global financial industry exceeding €7 billion annually. For this study, we operationalize market value using crowd-sourced estimates from Transfermarkt, a metric widely validated in sports economics as a robust proxy for actual transfer fees (Herm et al., 2014). While foundational research identifies performance metrics like goals as primary drivers (Frick, 2007), more recent scholarship highlights a "Superstar Effect" where non-performance factors like brand visibility magnify valuation (Franck & Nüesch, 2012; Rosen, 1981). However, the market has recently bifurcated into a pre-2020 hyper-inflationary era, termed the "Neymar Effect" (Poli et al., 2017), and a post-2020 pandemic correction (Drewes et al., 2021). This trajectory was violently interrupted by the COVID-19 pandemic, which caused an estimated €8.7 billion revenue shortfall across European football (Deloitte, 2021). Existing models typically treat players as static units nested within single clubs, ignoring the dynamic reality where players transfer between teams and accumulate value from both their portable human capital and the institutional prestige of their current employer. Neglecting this cross-classified structure likely misestimates the true drivers of market value in this volatile era.

To address these limitations, this study applies a cross-classified multilevel model to a decade of transfer data (2013–2024), centering the analysis at the 2020 pandemic baseline to quantify the impact of this structural break. By disentangling individual reputation from club context, we investigate four critical dynamics: the non-linear "Boom and Bust" market trajectory (RQ1)[1], the biological constraints of the player career arc (RQ2)[2], the valuation signal of "Squad Status" via relative playing time (RQ3)[3], and the existence of contextual premiums for club diversity or discounts for domestic talent in a globalized labor market (RQ4[4]-RQ5[5]).

2 Method

2.1 Sample and Variables

We obtained data from Transfermarkt and UEFA official rankings, both publicly available sources. We used purposive sampling to select the top 21 European clubs based on 10-year averaged UEFA coefficient rankings[6]. We then identified all players who played a full season in any of these clubs from 2013 through 2024 resulting in a sample of 7,023 player-season observations from 2,159 unique players across 21 clubs over 12 seasons. After listwise deletion of cases with missing citizenship data (approximately 2% loss), players averaged 26.5 years of age ($SD = 4.6$). Critically, 441 players (20.4%) played for multiple clubs during the study

[1] How has the transfer market evolved relative to the 2020 baseline? Specifically, does the market exhibit a non-linear "Boom and Bust" cycle that linear models miss?

[2] How does market valuation evolve across a player's biological career, and at what specific age does this value peak?

[3] To what degree does "Squad Status"—defined by relative playing time—predict market value independent of raw performance output?

[4] How is club-level international diversity associated with player valuations in a globalized labor market?

[5] How does the market valuation of domestic players compare to that of imported talent when controlling for performance and club context?

[6] UEFA coefficient rankings are a points-based system used by European soccer's governing body to rank national associations and clubs based on their performance in European competitions, which determines qualification spots and seeding for tournaments like the Champions League and Europa League.

period, justifying our cross-classified modeling approach.

The primary outcome variable is the natural logarithm of the player's inflation-adjusted market value (in 2024 USD). Predictors are categorized into three distinct domains: (1) Player Characteristics, including age, height, primary position (Attacker, Midfielder, Defender, Goalkeeper), and citizenship status relative to the club ("Homegrown" vs. Foreign); (2) Performance and Status, captured by goals, assists, disciplinary records (red/yellow cards), and playing time; and (3) Club Context, specifically the proportion of foreign players on the roster and the club's UEFA coefficient, which serves as a proxy for recent competitive success and prestige. For further descriptive statistics on each variable see Appendix A, Table1.

2.2 Cross-Classified Data Structure

Our data exhibit a cross-classified structure where players (i) are constant units, but club membership (j) changes over time (t). When a player transfers from club j_1 to club j_2, club-level predictors update to reflect the new club's characteristics. This structure requires modeling player and club as crossed random effects rather than nested effects. Figure 1 is a visual of a cross-classified model example of three players that either move through clubs or remain in the same club throughout their career.

Figure 1. Cross−classified Data Structure Highlights How Players Move Between Clubs Ov

2.3 Model Specification

We specify a cross-classified multilevel model to partition the variance in market value across observations (t), players (i), and clubs (j). Crucially, because club attributes like internationalization and UEFA ranking vary year-to-year, they are modeled as Level 1 (time-varying) predictors rather than static Level 2 traits.

Level 1: Observation Model (Time-Varying Effects)

At the observation level, the log-transformed market value is modeled as a function of the player-club intercept (π_{0ij}), algebraic time trends, biological aging, and dynamic contextual metrics:

$$\begin{aligned}\log(\mathrm{MV})_{tij} = \pi_{0ij} \\ &+ \pi_{1j}(\mathrm{Year}_t) + \pi_{2j}(\mathrm{Year}_t^2) + \pi_3(\mathrm{Year}_t^3) \\ &+ \pi_4(\mathrm{Age}_{it} - \bar{A}) + \pi_5(\mathrm{Age}_{it} - \bar{A})^2 \\ &+ \pi_6(\mathrm{Mins}_{Z,it}) \\ &+ \pi_7(\mathrm{Foreign}_{tj}) + \pi_8(\mathrm{UEFA}_{tj}) \\ &+ {}_{perf} + \epsilon_{tij}\end{aligned}$$

Note: MV = Market Value. The subscripts tj for Foreign and UEFA denote that these club attributes vary by time t.

Level 2a: Player Model (Time-Invariant Traits)

The player-club intercept (π_{0ij}) is decomposed into a fixed component based on static player characteristics and a random player effect (u_{0i}):

$$\pi_{0ij} = \beta_{0j} + \gamma_1(\mathrm{Height}_i) + \gamma_2(\mathrm{Citizen}_{ij}) + {}_{demog} + u_{0i}$$

Level 2b: Club Model (Contextual Trajectories)

The club baseline (β_{0j}) and the time-varying slope parameters (π_{1j}, π_{2j}) include random effects to allow for heterogeneity in both growth velocity and trajectory curvature:

$$\begin{aligned}
\beta_{0j} &= \delta_{00} + v_{0j} && \text{(Baseline Prestige)} \\
\pi_{1j} &= \delta_{10} + v_{1j} && \text{(Linear Growth Deviation)} \\
\pi_{2j} &= \delta_{20} + v_{2j} && \text{(Quadratic Curvature Deviation)} \\
\pi_{3} &= \delta_{30} && \text{(Fixed Cubic Trend)}
\end{aligned}$$

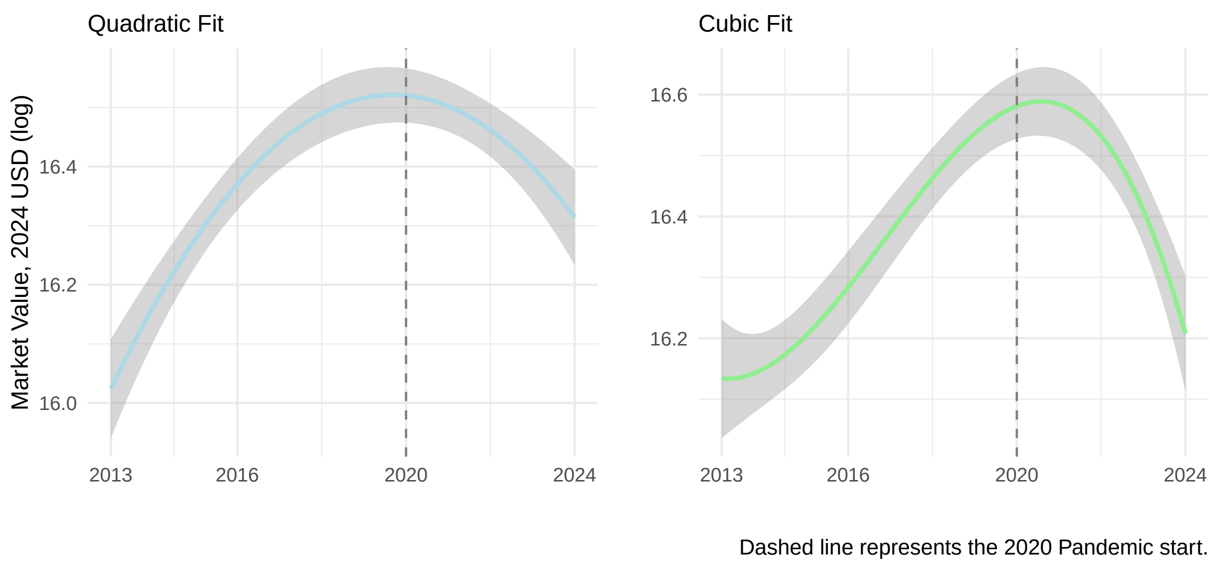
Variance-Covariance Structure

$$u_{0i} \sim N(0, \sigma_u^2) \quad , \quad \begin{pmatrix} v_{0j} \\ v_{1j} \\ v_{2j} \end{pmatrix} \sim N\left(\mathbf{0}, \begin{pmatrix} \sigma_{Int}^2 & \sigma_{01} & \sigma_{02} \\ \sigma_{01} & \sigma_{Lin}^2 & \sigma_{12} \\ \sigma_{02} & \sigma_{12} & \sigma_{Quad}^2 \end{pmatrix} \right)$$

2.4 Variables and Centering Strategies

To facilitate model convergence and ensure substantively meaningful intercepts, we applied specific centering and transformation strategies. Most notably, `Season Year` was centered at the 2020 pandemic baseline (2020=0), allowing us to interpret the intercept as the expected market value at the onset of the COVID-19 structural break. We introduced polynomial terms to capture theorized non-linear dynamics (see Figure 2): a cubic term for time to model the "Boom-and-Bust" market cycle (pre-pandemic inflation followed by post-pandemic deceleration), and a quadratic term for `Age` to empirically test the biological "Inverted-U" career trajectory. Additionally, `Minutes Played` was standardized (Z-scored) relative to each specific club-year to isolate a player's hierarchical status (e.g., Starter vs. Bench) independent of team-level rotation policies, while performance metrics were centered by position to account for role-specific expectations.

Figure 2: Market Value Impacted by Pandemic



2.5 Model Building and Statistical Software

We employed sequential model building: (1) null model for ICC estimation; (2) linear time polynomial; (3) cubic time polynomial; (4) random slope for quadratic time at club level[7]; (5) player/club-season Level 1 predictors with quadratic age. Model selection was guided by likelihood ratio tests and information criteria. Model selection and comparison was guided by Likelihood Ratio Tests (LRT) to evaluate the statistical significance of nested model improvements. To balance goodness-of-fit against model complexity, we also consulted the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). Models yielding significant LRT results ($p<.05$) and lower values on both information criteria were retained for subsequent analysis. We describe each model in Appendix 2, Table 2 alongside ICC, AIC, and BIC values.

All analyses were conducted in R version 4.5.1 using the `lme4 1.1-37` package for mixed-

[7]We excluded player-level random slopes due to data sparsity; a majority of players appear in the dataset for only one or two seasons, making it mathematically impossible to disentangle their individual growth trajectories from residual error. In contrast, the club-level data provides a robust longitudinal structure with continuous observations across the decade, allowing for the stable estimation of club-specific market value trajectories.

effects model estimation with a Restricted Maximum Likelihood (REML) method. REML was selected to provide unbiased estimates of the variance components (random effects) by accounting for the degrees of freedom lost when estimating the fixed effects, ensuring that club and player heterogeneity were not underestimated.. To ensure robust model convergence given the complexity of the random effects structure, we utilized the BOBYQA (Bound Optimization BY Quadratic Approximation) optimizer. This algorithm is specifically designed to handle boundary constraints (such as variance components approaching zero) more effectively than the default Nelder-Mead optimizer. . Significance tests used Satterthwaite approximations via the `lmerTest 3.1-3` package.

3 Results

The unconditional means (null) model revealed substantial clustering at the player level. Player-level variance accounted for 78% of the total variance ($ICC_{player} = .75$), while club-level variance accounted for 5% ($ICC_{club} = .03$), highlighting that persistent individual heterogeneity dominates market valuations. Regarding temporal dynamics, model comparisons favored a cubic fixed effect for season year over a linear specification, $\chi^2(2) = 45.66, p < .001$. We subsequently tested for club-specific heterogeneity in growth trajectories. A likelihood ratio test using a 50:50 mixture of χ^2_1 and χ^2_2 distributions[8] confirmed that adding a random linear slope significantly improved model fit compared to the random-intercept model, $\chi^2(5) = 97.13, p < .001$, indicating that clubs differ significantly not only in their growth rates but also in the curvature of their market value trajectories.[9] Consequently, our final best-fitting model incorporates these complex random effects alongside time-varying player

[8]We test whether a random slope with quadratic season year at the club level has better fit to the data. We find evidence supporting the need to include these random slopes using a likelihood ratio test with an alternative hypothesis, $H_A = \tau_{11} > 0$ since variance should be larger than 0. To test this hypothesis we use the likelihood ratio test (LRT) simply by $-2[\log \text{likelihood}_{reduced} - \log \text{likelihood}_{full}]$ to examine the difference. This code can be found in Appendix B.

[9]While we included random linear and quadratic slopes, a random cubic slope was excluded due to over-parameterization. Estimating the required 10 variance parameters with only 21 clubs led to singularity; therefore, the quadratic specification represents the maximal complexity supported by the data.

and club predictors ($\chi^2(19) = 5626, p < .001$ compared to random slopes model). Model comparisons are shown in Appendix C, Table 1. Here we output our final model's variance components and fixed effects.

3.1 Model Diagnostics

We evaluated the statistical assumptions underlying our cross-classified multilevel model through visual and quantitative diagnostics. The model converged successfully using the BOBYQA optimizer with no warnings, indicating stable estimation of all variance components. We examined the normality of random effects at both levels via quantile-quantile (Q-Q) plots. The player-level random intercepts ($n = 2,159$) displayed excellent adherence to normality across the entire distribution, consistent with our large sample providing stable estimates of individual heterogeneity. The club-level random effects ($n = 21$) showed minor departures from normality in the extreme tails, which is expected given the small number of Level 2 units; however, simulation studies demonstrate that multilevel models remain robust to such departures when the sample sizes at Level 1 are large (Maas & Hox, 2004). The residual versus fitted values plot revealed homoscedastic variance across the range of predicted values, with no systematic patterns suggesting model misspecification. We also assessed multicollinearity among the fixed effects using variance inflation factors (VIF); all values remained below 2.5, indicating no problematic collinearity. Full diagnostic plots are presented in Appendix C.

Table 1. Estimated Variance Components for Final Model

Component	Variance	SD
Player (Reputation)	0.9254	0.9620
Club (Baseline Prestige)	0.0541	0.2326
Club (Linear Growth)	0.0105	0.1025
Club (Quadratic Curvature)	0.0073	0.0855

Residual (Unexplained)	0.2070	0.4550

Table 2. Fixed Effects Estimates from Final Model

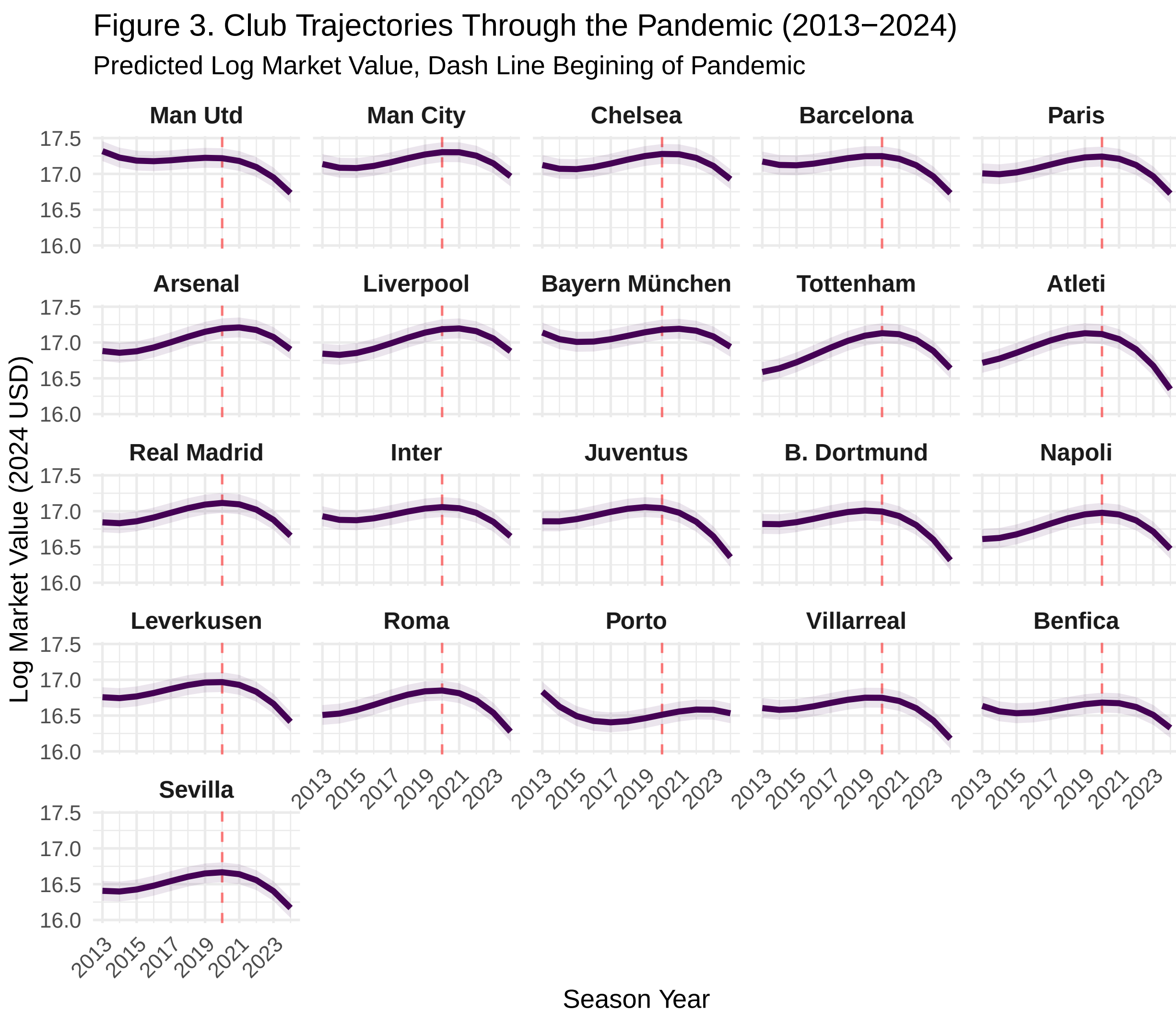
Predictor		SE	t	% Change	
Year (linear)	0.002	0.030	0.08	0.2	
Year²	-0.270	0.024	-11.05	-23.7	***
Year³	-0.117	0.011	-10.98	-11.0	***
Height	0.010	0.004	2.81	1.0	**
Defender	-0.403	0.058	-6.97	-33.2	***
Goalkeeper	-1.059	0.095	-11.17	-65.3	***
Midfielder	-0.190	0.058	-3.29	-17.3	**
Domestic citizen	-0.185	0.031	-5.89	-16.9	***
Age (linear)	0.019	0.004	5.02	1.9	***
Age²	-0.024	0.000	-63.08	-2.4	***
goals	0.005	0.002	2.36	0.5	*
assists	0.010	0.003	3.51	1.0	***
Red cards	0.009	0.022	0.42	0.9	
Yellow cards	-0.002	0.003	-0.49	-0.2	
Minutes (Z)	0.350	0.013	26.20	41.9	***
UEFA Coefficient	0.000	0.001	-0.39	0.0	
Proportion of Foregin Players	0.103	0.104	0.99	10.8	

Note: *** p < .001, ** p < .01, * p < .05

3.1.1 Market Evolution

Using raw polynomials centered at 2020 ($t = 0$), our model mathematically pinpoints the moment the market froze. The linear coefficient at the 2020 baseline is effectively zero ($\pi_1 = 0.00, p = 0.93$), indicating that at the exact onset of the pandemic, the relentless inflation of the previous decade came to a complete halt. The significant negative quadratic ($\pi_2 =$

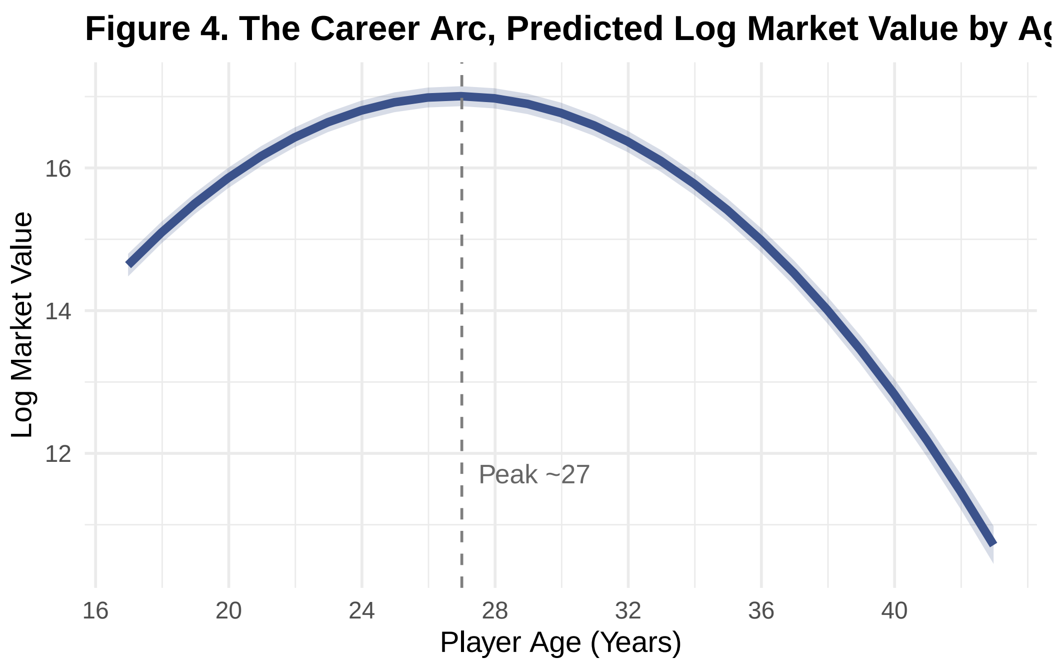
-0.27) and cubic ($\pi_3 = -0.12$) terms confirm the transition from a pre-2019 "Boom" into a structural correction. Figure 3 visualizes the complexity of club-level market dynamics by plotting the predicted trajectory for all clubs individually. This faceted view highlights how clubs diverge from the global trend. The vertical position of each chart confirms the initial hierarchy (e.g., random Intercepts), with elite clubs like Manchester City and Bayern Munich maintaining a consistently higher baseline than selling clubs (e.g., Sevilla, Porto). The distinct curvature of each line, particularly around the 2020 pandemic baseline, illustrates the significant variance in the random slopes. Clubs with concave trajectories (e.g., Juventus, Barcelona) plateaued or declined rapidly, confirming the risk of stagnation. Conversely, clubs with convex trajectories (e.g., Arsenal, Liverpool) show stronger momentum, indicating a more resilient recovery and active value generation throughout the decade.



3.1.2 Biological Age

Our model predicts the peak age to be 26.9[10] for optimal transfer market value we visualize in Figure 4 the biological constraints on valuation. The curve rises steeply as players transition from 'prospects' (Age 17–23), flattens into a 'prime' plateau between ages 26 and 28, and declines steadily thereafter. The model predicts that a player's value drops by approximately 1.5 log points (from a peak of 17.0 to 15.5) as they age from 27 to 35, reflecting the market's heavy discount on older assets with limited resale potential.

[10]We use the delta method, specifically Taylor series approximation, to propagate uncertainty through nonlinear transformation of $-\beta_{age}/(2 \times \beta_{age^2})$.

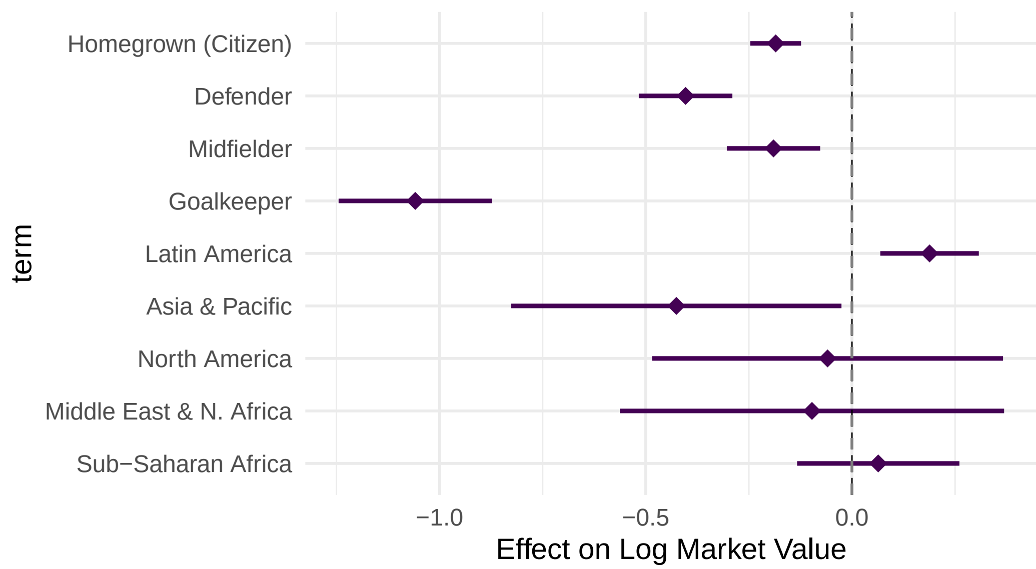


3.1.3 Contextual Premiums and Discounts

Squad status remains the dominant behavioral driver. The standardized coefficient for minutes played ($\pi_6 = 0.35$) indicates a 42% valuation premium ($e^{0.35} - 1$) for key starters (+1 SD) compared to average players. Conversely, we observe a robust "Domestic Discount," where homegrown citizens are valued approximately 17% lower ($\gamma_2 = -0.19$) than comparable foreign imports. While we hypothesized a "Diversity Premium" for clubs with higher international representation, the fixed effect for `foreign_players` was positive but not statistically significant ($\pi_7 = 0.10, p = 0.32$) in the final model. Similarly, the `uefa_coefficient` showed no significant effect ($p = 0.70$). This suggests that once we account for the massive heterogeneity in Club Random Effects (i.e., Baseline Prestige and Growth Trajectories) and individual player reputation ($ICC \approx 0.78$), specific club-level metrics add little unique explanatory power. In Figure 5, we present relative market valuation for Level 2 fixed effects (e.g., citizenship, region origin, and position). The coefficient for "Homegrown (Citizen)" falls strictly to the left of zero, confirming that domestic players are valued significantly lower than comparable foreign imports ($\beta = -0.19$), potentially reflecting the recruitment premium required to attract international talent. Attackers (the baseline) command the highest market value. All other positions show significant negative effects, with Goalkeepers facing the steepest penalty ($\beta = -1.06$), suggesting the market disproportionately rewards offensive output. While most regional coefficients overlap with zero (i.e., indicating no difference from the European baseline), Latin America stands out as the only region with a positive premium ($\beta = 0.19$). This suggests that South American players function as a "premium export brand" in the global market, distinct from other non-European origins.

Figure 5. Valuation Gaps Across Citizenship,

Discounts (Left) vs. Premiums (Right) relative to baseline



4 Discussion

This study demonstrates that player market value in elite football is driven primarily by persistent individual reputation but is meaningfully moderated by club context and market cycles. Our cross-classified approach properly attributes variance to portable player effects versus transient club effects. In this section, we synthesize our findings, articulate their theoretical contributions, and outline practical implications for stakeholders in the transfer market.

4.1 Theoretical Contributions

Our findings advance the literature on football economics in several important ways.

The Pandemic as a Structural Break. Prior research has documented general inflation in player valuations (Poli et al., 2017), but few studies have rigorously tested for structural breaks in market dynamics. By centering our temporal polynomial at 2020, we provide direct evidence that the COVID-19 pandemic represented a fundamental discontinuity in the transfer market. The near-zero linear slope at this inflection point indicates that the decade-long inflationary trend abruptly halted, followed by a period of market correction captured by our negative quadratic and cubic terms. This finding suggests that external macroeconomic shocks can fundamentally alter the valuation calculus in professional sports.

Biological Determinism in Valuation. The strict inverted-U relationship between age and market value, with a peak at approximately 27 years, extends the career arc literature to transfer market contexts. Unlike performance metrics that may plateau, market value appears to follow a more deterministic biological trajectory. The steepness of the post-peak decline (approximately 1.5 log points from age 27 to 35) reflects the market's forward-looking assessment of remaining productive years and resale potential. This finding underscores that transfer markets price both current ability and future option value.

Squad Status as a Market Signal. The dominance of minutes played ($\beta = 0.35$) over position-specific performance metrics (goals, assists) suggests that the market relies heavily on managerial assessments as signals of underlying quality. This finding aligns with signaling theory, where coaches possess private information about player quality that is revealed through squad selection decisions. The standardized coefficient indicates that a one standard deviation increase in relative playing time yields a 42% valuation premium, representing the single largest behavioral effect in our model.

The Domestic Discount Paradox. Our finding that homegrown players are valued approximately 17% lower than comparable foreign imports presents a puzzle for economic theory. While labor market friction explanations would predict premiums for mobile workers who bear relocation costs, our results suggest an alternative mechanism: domestic players may be systematically undervalued because their availability is taken for granted, whereas international recruitment requires competitive bidding that inflates prices. This "scarcity premium" for foreign talent represents a potential market inefficiency that astute clubs could exploit.

4.2 Practical Implications

Our findings carry actionable implications for multiple stakeholders in the football ecosystem.

For Buying Clubs. The identified market inefficiencies suggest several value-seeking strategies. First, targeting domestic players may offer significant cost savings without sacrificing quality, as the 17% domestic discount appears unrelated to on-field performance. Second, the strong age effect implies that players approaching 27 represent peak value-for-money, as they offer immediate prime-year performance before the steep depreciation curve. Third, clubs can leverage our model to identify "undervalued" players whose market price falls below their predicted value based on observable characteristics.

For Selling Clubs. The temporal dynamics reveal optimal timing strategies for player sales. The market correction following 2020 suggests that selling clubs should be attentive to macroeconomic conditions and broadcast revenue cycles when timing major sales. Additionally, the steep post-peak age depreciation argues for selling established players earlier rather than later, particularly those entering their late twenties. Clubs that delay sales risk substantial value erosion as players age beyond their peak market window.

For Agents and Players. Individual career planning can benefit from understanding the structural determinants of market value. Players seeking to maximize their market value should prioritize securing starting positions (maximizing minutes played) over accumulating individual statistics, as squad status carries greater market weight. Furthermore, the biological constraints on valuation suggest that contract negotiations should front-load compensation for players approaching age 27, recognizing that subsequent contracts will occur on the declining portion of the age-value curve.

4.3 Key Findings Summary

We summarize our substantive findings organized by research question:

Market Cycles. The cubic temporal trend reveals distinct boom-bust phases rather than constant market growth. The rapid inflation period (2015-2019) following the Neymar transfer record and subsequent post-pandemic correction highlight the cyclical nature of transfer market valuations.

Biological Constraints. The strict inverted-U age relationship, with peak value around age 27, confirms that physical peak represents the optimal balance between experience and athleticism, with practical implications for optimal transfer timing.

Squad Status Dominance. Minutes played emerged as the strongest predictor, with starters valued over 40% higher than bench players. This universal signal transcends position-specific metrics, suggesting that coaches' implicit assessments are incorporated into market valuations.

Contextual Premiums. The domestic discount for homegrown players suggests systematic biases that could be exploited for competitive advantage.

4.4 Limitations and Future Directions

Several limitations warrant acknowledgment. First, our sample focuses exclusively on elite Big Five league clubs, limiting generalizability to lower divisions or emerging football markets where different valuation dynamics may operate. Second, minutes played is likely endogenous to market value; highly valued players may receive more playing time precisely because of their status, meaning our estimates should be interpreted as associations rather than causal effects. Future research could employ instrumental variable approaches, such as using injury-induced absences or managerial changes as exogenous variation in playing time. Third, we constrained the cubic time polynomial to be fixed at the club level due to convergence issues with higher-order random slopes; additional data or Bayesian estimation approaches may permit more flexible specifications. Fourth, residual diagnostics revealed minor departures from normality in the extreme tails, though our large sample size ensures robust estimation under asymptotic theory.

Several promising directions emerge for future research. First, extending our cross-classified framework to lower-tier leagues would test whether the identified mechanisms generalize beyond elite contexts. Second, incorporating more granular performance data (e.g., expected goals, pressing statistics) could reveal which specific on-field contributions drive valuations beyond aggregate metrics. Third, examining how club financial health interacts with player valuation could illuminate whether distressed clubs systematically undervalue their assets. Finally, quasi-experimental designs exploiting policy changes (e.g., Brexit's impact on non-EU player quotas) could strengthen causal claims about contextual premiums and discounts.

4.5 Conclusion

By properly modeling the cross-classified structure where players transfer between clubs, we quantify how individual performance, biological constraints, and club context jointly shape valuations in elite European football. The substantial player-level variance ($ICC = 0.78$) confirms that persistent individual reputation dominates market assessments, while club context creates meaningful premiums and growth trajectories. Our findings reveal that the transfer market, while responsive to observable performance metrics, also incorporates forward-looking assessments of biological decline, managerial signals through squad selection, and contextual factors that create systematic premiums and discounts. These patterns represent both economic regularities that clubs can anticipate and potential inefficiencies that astute market participants can exploit. As the football transfer market continues to evolve in scale and sophistication, rigorous multilevel modeling approaches offer a principled framework for understanding the complex interplay of individual talent, institutional context, and market dynamics.

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5 Appendix A

Table 1: Variable Descriptions and Descriptive Statistics (N=7,168 raw)

Description	Raw Stats (Mean/SD)	Transformation Strategy
Continuous Variables		
Market Value (2024 USD)	$25.8M (29.0M)	*Log-transformed (ln)*
Season Year	2018.63 (3.47)	*Centered at 2020 (0 = 2020)*
Player Age (Years)	26.50 (4.58)	*Grand-mean centered + Quadratic*
Height (cm)	182.10 (6.84)	*Grand-mean centered*
Minutes Played	1761.77 (1337.70)	*Z-scored within Club-Year*
Goals Scored	3.41 (6.25)	*Centered by Position Mean*
Assists Made	2.79 (4.00)	*Centered by Position Mean*
Club Foreign %	63.5% (11.1%)	*Grand-mean centered*
UEFA Coefficient	20.42 (6.89)	*Year-mean centered*
Red Cards	0.07 (0.28)	*Raw count*
Yellow Cards	3.42 (3.60)	*Raw count*
Categorical Variables		
Citizenship (Homegrown)		*Binary (1=Citizen)*
- No	4498 (62.8%)	
- Yes	2567 (35.8%)	
- NA	103 (1.4%)	
Primary Position		*Factor (Ref: Attack)*
- Attack	1925 (26.9%)	
- Defender	2448 (34.2%)	
- Goalkeeper	605 (8.4%)	
- Midfield	2190 (30.6%)	
Region of Origin		*Factor (Ref: Europe)*
- East Asia & Pacific	69 (1.0%)	
- Europe & Central Asia	5139 (71.7%)	
- Latin America & Caribbean	1441 (20.1%)	
- Middle East & North Africa	56 (0.8%)	
- North America	52 (0.7%)	
- Sub-Saharan Africa	303 (4.2%)	
- NA	108 (1.5%)	

6 Appendix B

6.1 Annotated Model Syntax

```
#______________________________________________________________________________
# MODEL SPECIFICATION & FITTING
#______________________________________________________________________________

# 1. MODEL NULL (Unconditional Means)
# Purpose: Partition variance (calculate ICC) between Player and Club levels.
model_null <- lmer(usd_revenue_2024_log ~ (1|player_id) + (1|club_id),
                   data = clean_player_df, REML = TRUE, control = strict_control)

# 2. MODEL LINEAR TIME TREND
# Purpose: Test for general linear inflation in market values
## (Base model for time).
model_time_linear <- lmer(usd_revenue_2024_log ~ season_year_c +
                          (1|player_id) + (1|club_id),
                          data = clean_player_df, REML = TRUE, control = strict_control)

# 3. MODEL CUBIC TIME TREND (Fixed Effects)
# Purpose: Test for non-linear "Boom and Bust" cycles.
## (raw=TRUE for interpretability)
model_time_cubic <- lmer(usd_revenue_2024_log ~ poly(season_year_c, 3, raw=TRUE) +
                         (1|player_id) + (1|club_id),
                         data = clean_player_df, REML = TRUE, control = strict_control)
```

```
# 4. MODEL RANDOM QUADRATIC SLOPE (Club Level)
# Purpose: Test if trajectory curvature varies by club.
## (Linear slope model omitted for brevity)
model_slope_quadratic <- lmer(usd_revenue_2024_log ~ poly(season_year_c, 3, raw=TRUE) +
                              (1|player_id) + (poly(season_year_c, 2, raw=TRUE) | club_i
                              data = clean_player_df, REML = TRUE, control = strict_cont

# 5. MODEL (FINAL): FULL SPECIFICATION
# Purpose: Estimate marginal effects of performance, status,
## and context controlling for time.
model_player <- lmer(
  usd_revenue_2024_log ~
    poly(season_year_c, 3, raw=TRUE) +          # Temporal Controls
    poly(age_c, 2, raw=TRUE) +                  # Career Arc
    height_c + position + Region +              # Player Traits
    citizen_of_club_country +                   # Context
    goals + assists + red_cards + yellow_cards + minutes_played + # Performance
    uefa_coefficient + foreign_players +        # club fixed effects
    (1|player_id) + (poly(season_year_c, 2, raw=TRUE) | club_id), # Random Effects
  data = clean_player_df, REML = TRUE, control = strict_control
)

#_______________________________________________________________________________
# LIKELIHOOD RATIO TEST (MIXTURE DISTRIBUTION)
#_______________________________________________________________________________
# Corrects p-values for testing variance components on the boundary (Stram & Lee, 1994)
# LRT test for 2 random slopes
```

```r
calculate_lrt_mixture <- function(reduced_model, full_model) {

  # 1. Extract Log-Likelihoods
  ll_reduced = as.numeric(logLik(reduced_model))
  ll_full = as.numeric(logLik(full_model))

  # 2. Calculate LRT Statistic
  lrt_stat = -2 * (ll_reduced - ll_full)

  # 3. Calculate Difference in Parameters (df)
  df_reduced = attr(logLik(reduced_model), "df")
  df_full = attr(logLik(full_model), "df")
  df_diff = df_full - df_reduced

  # 4. Calculate P-Value based on the difference

  if (df_diff == 3) {
    # --- NEW BLOCK FOR QUADRATIC SLOPE ---
    # Going from 2 Random Effects (Int+Lin) -> 3 Random Effects (Int+Lin+Quad)
    # We add 1 variance + 2 covariances = 3 parameters.
    # The mixture is 0.5 * ChiSq(2) + 0.5 * ChiSq(3)
    p_val = 0.5 * pchisq(lrt_stat, df = 2, lower.tail = FALSE) +
      0.5 * pchisq(lrt_stat, df = 3, lower.tail = FALSE)

    msg = "Mixture of ChiSq(2) and ChiSq(3)"

  } else if (df_diff == 2) {
```

```
    # Going from 1 Random Effect (Int) -> 2 Random Effects (Int+Lin)
    # Adds 1 variance + 1 covariance = 2 parameters
    p_val = 0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE) +
      0.5 * pchisq(lrt_stat, df = 2, lower.tail = FALSE)

    msg = "Mixture of ChiSq(1) and ChiSq(2)"

  } else if (df_diff == 1) {
    # Adding a single parameter (e.g. just a variance, no covariance)
    p_val = 0.5 * pchisq(lrt_stat, df = 0, lower.tail = FALSE) +
      0.5 * pchisq(lrt_stat, df = 1, lower.tail = FALSE)

    msg = "Mixture of ChiSq(0) and ChiSq(1)"

  } else {
    # Fallback for standard fixed effects comparisons
    p_val = pchisq(lrt_stat, df = df_diff, lower.tail = FALSE)
    msg = paste0("Standard ChiSq(", df_diff, ")")
  }

  # Output
  cat("LRT Statistic:", round(lrt_stat, 4), "\n")
  cat("Parameter Diff:", df_diff, "\n")
  cat("Distribution:  ", msg, "\n")
  cat("P-Value:       ", format.pval(p_val, eps = .0001), "\n")
}
```

```
LRT Statistic: 97.1306

Parameter Diff: 5

Distribution:   Standard ChiSq(5)

P-Value:        < 0.0001
```

7 Appendix C

Table 1. Model Fit Comparison (AIC, BIC, and ICC)

Models	**Purpose & Research Justification**	ICCplayer/club	AIC	BIC
Null	Calculate ICC and partition variance between players and clubs before adding predictors	.75/.03	20105.19	20132.62
Time Linear	Test for a general linear trend in market value growth	.76/.03	20050.17	20084.46
Time Cubic	Test for non-linear market dynamics (e.g., acceleration or deceleration/plateaus)	.76/.03	20021.61	20069.61
Random Slope Quadratic	Test if the growth curvature varies by club	.76/.04	19934.48	20016.76
Varying Level 1 fixed effects	Estimate the marginal value of performance metrics and player/club characteristics controlling for time	.78/.05	14467.31	14679.87
Final Model Selection	Compare models using LRT, AIC/BIC, and residual diagnostics (QQ plots) to select the most parsimonious and robust fit			

Figure 6. Model Diagnostic Plots

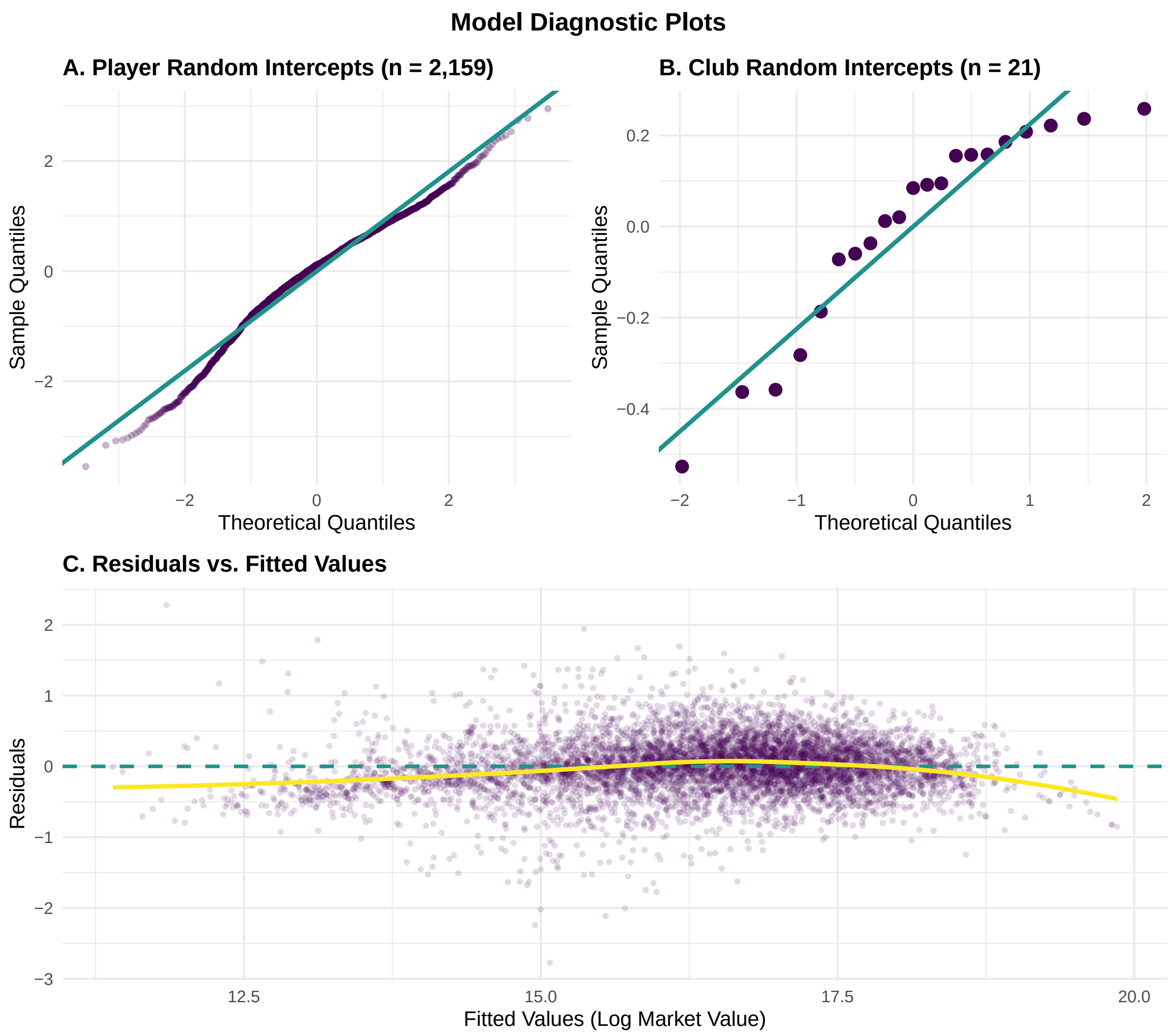

Figure 1: Diagnostic plots for the final cross-classified multilevel model. Panel A shows the Q-Q plot for player random intercepts (n=2,159), Panel B shows the Q-Q plot for club random intercepts (n=21), and Panel C displays residuals versus fitted values to assess homoscedasticity.

Figure 7. Club Random Effects Correlation Structure

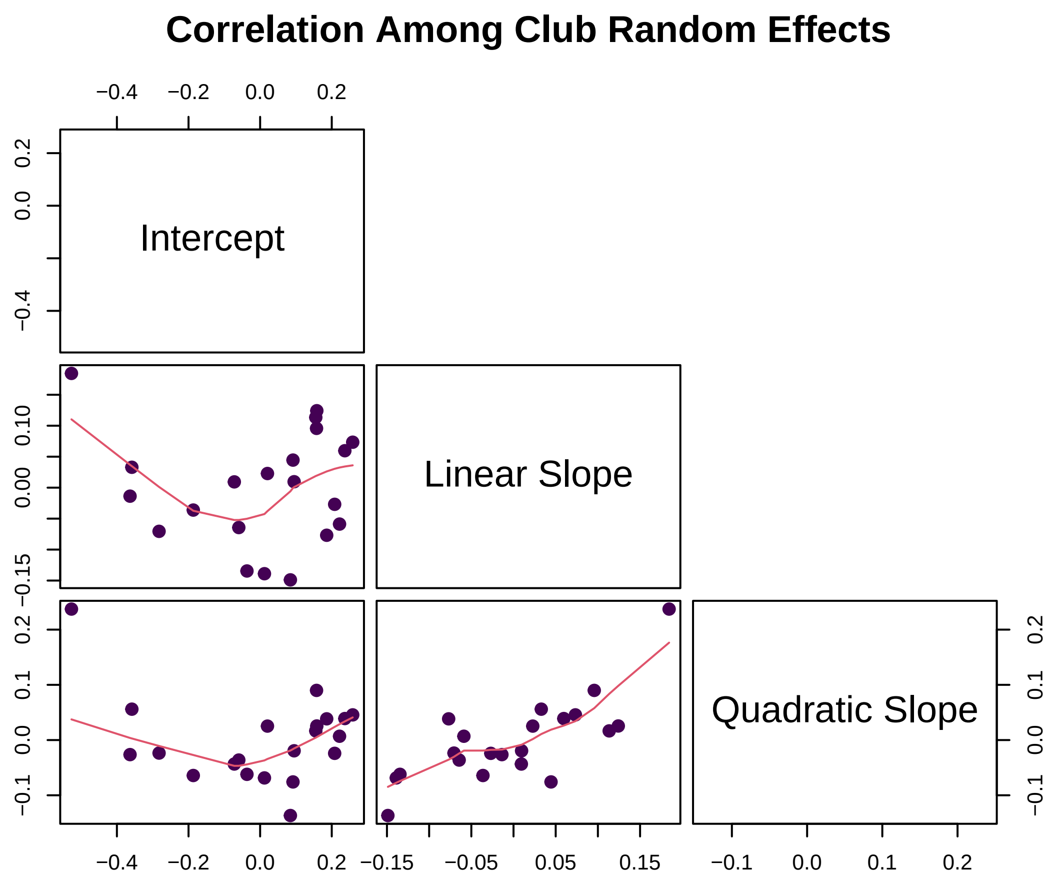

Figure 2: Scatterplot matrix showing correlations among club-level random effects (intercept, linear slope, and quadratic slope). Strong correlations suggest that clubs with higher baseline prestige also exhibit distinct growth trajectories.